

Umbrella Sampling Simulations Measure Switch Peptide Binding and Hydrophobic Patch Opening Free Energies in Cardiac Troponin

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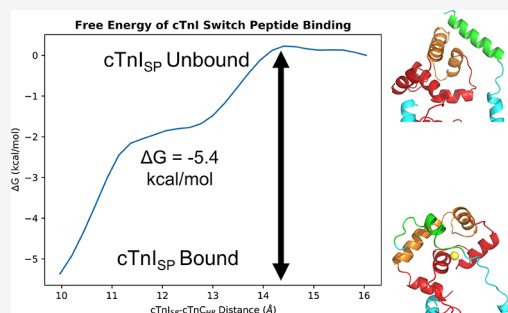


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ABSTRACT: The cardiac troponin (cTn) complex is an important regulatory protein in heart contraction. Upon binding of Ca^{2+} , cTn undergoes a conformational shift that allows the troponin I switch peptide (cTnI_{SP}) to be released from the actin filament and bind to the troponin C hydrophobic patch (cTnC_{HP}). Mutations and modifications to this complex can change its sensitivity to Ca^{2+} and alter the energetics of the transition from the Ca^{2+} -unbound, cTnI_{SP}-unbound form to the Ca^{2+} -bound, cTnI_{SP}-bound form. We utilized targeted molecular dynamics (TMD) to obtain a trajectory of this transition pathway, followed by umbrella sampling to estimate the free energy associated with the cTnI_{SP}–cTnC_{HP} binding and the cTnC_{HP} opening events for wild-type (WT) cTn. We were able to reproduce experimental values for the cTnI_{SP}–cTnC_{HP} binding event and obtain cTnC_{HP} opening free energies in agreement with previous computational measurements of smaller cTnC systems. This excellent agreement for WT cTn demonstrated the strength of computational methods in studying the dynamics and energetics of the cTn complex. We then introduced mutations to the cTn complex that cause cardiomyopathy or alter its Ca^{2+} sensitivity and observed a general decrease in the free energy of opening the cTnC_{HP}. For these same mutations, we observed no general trend in the effect on the cTnI_{SP}–cTnC_{HP} binding event. Our method sets the stage for future computational studies on this system that predict the consequences of yet uncharacterized mutations on cTn dynamics and energetics.



INTRODUCTION

Contraction of the thin filament in the heart muscle is regulated by Ca^{2+} binding to the cardiac isoform of the troponin (cTn) complex.¹ Upon binding of Ca^{2+} to cTn, a conformational shift occurs that releases cTn from actin, exposing myosin binding sites that allow muscle contraction to occur. The exact mechanism of the cTn conformational transition and how cTn mutations affect this process is still not fully understood.

The cTn complex is comprised of three subunits: the Ca^{2+} -binding subunit (cTnC), the inhibitory subunit (cTnI), and the tropomyosin-binding structural subunit (cTnT) (Figure 1). cTnC can be divided in two regions: N-terminal cTnC (NcTnC, residues 1–89) contains Ca^{2+} binding sites I and II, and C-terminal cTnC (CcTnC, residues 90–161) contains Ca^{2+} binding sites III and IV. Under normal physiological conditions, sites III and IV are always occupied, and site I is never occupied by Ca^{2+} . Thus, site II of NcTnC is predominately responsible for sensing Ca^{2+} for muscle contraction. When Ca^{2+} binds to site II, it induces a conformational and dynamical shift that exposes a hydrophobic patch (cTnC_{HP}) in NcTnC that is able to bind the cTnI switch peptide (cTnI_{SP}, residues 149–164). The cTnI_{SP} (along with the cTnI inhibitory peptide) interacts with actin in the Ca^{2+} -unbound state, and its dissociation from actin and subsequent binding to the cTnC_{HP} in the cTn Ca^{2+} -bound state allow for

the thin filament interaction with myosin necessary for muscle contraction.

Mutations throughout the cTn complex subunits have been shown to cause cardiomyopathy, a disease that can lead to heart failure or sudden cardiac death.² Three types of cardiomyopathy caused by these mutations include hypertrophic (HCM), restrictive (RCM), and dilated (DCM). HCM causes thickening of the interventricular septum, leading to decreased ventricular filling and impaired diastolic function of the heart. RCM is a rarer disease but causes similar problems with decreased ventricular filling without thickening of the heart muscle. Both HCM and RCM mutations have been shown to increase Ca^{2+} sensitivity in cTnC. DCM has been shown to decrease cTnC Ca^{2+} sensitivity, with physiological effects including an increase in the size of the left ventricle, or both ventricles, and impaired systolic function.³

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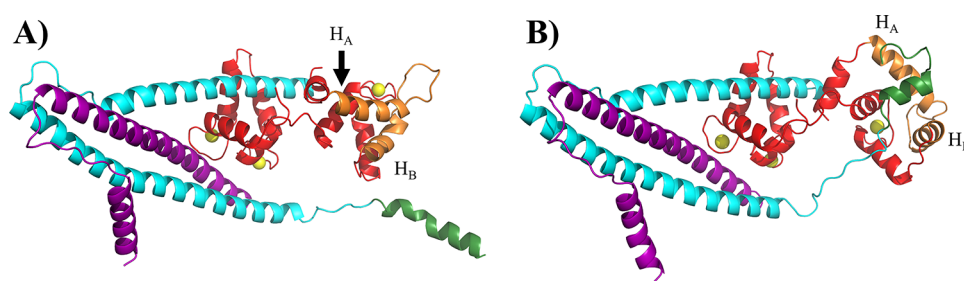


Figure 1. Cartoon representation of the cTn complex. A) TMD starting model representing Ca^{2+} -unbound, cTnI_{SP}-unbound cTn (PDB: 6KN7, chains a, b, and c). B) TMD final model representing Ca^{2+} -bound, cTnI_{SP}-bound cTn (PDB: 6KN8, chains a, b, and c). cTnT is depicted in purple, cTnI is cyan, and cTnC is red. Helices A and B (residues 14–48, cTnC_{HP}) of cTnC are shown in orange and labeled H_A and H_B, respectively. The cTnI_{SP} (cTnI residues 149–164) are shown in green. Ca^{2+} ions added by AutoDock Vina are yellow.

Structurally, cTn has been well characterized. A partial crystal structure of the Ca^{2+} -bound cTn complex, published by Takeda et al.⁴ in 2003, constituted a major advancement of our structural understanding of the complex. More recently, a cryo-EM structure of the entire cTn complex in the context of the thin filament was published by Yamada et al.⁵ This study was able to elucidate structures of the cTn complex, anchored onto the thin filament, in the Ca^{2+} -bound and Ca^{2+} -unbound states. Although this breakthrough was able to provide snapshots of the cTn complex and thin filament in both states, it was unable to provide information on the conformational transition and thermodynamic pathway between the two. A quantitative assessment of these processes is desirable for our understanding of heart muscle biology and disease, drug discovery, and protein design.

Computational methods can facilitate the study of these processes.⁶ Previous efforts have been able to utilize molecular dynamics (MD) and free energy methods to study Ca^{2+} binding to cTnC site II.^{7–11} MD methods have also been able to investigate disruptions in dynamics upon the introduction of mutations that alter site II's Ca^{2+} sensitivity or mutations that have been linked with disease.^{12–14} Enhanced sampling methods, umbrella sampling (US), and long timescale simulations have been utilized to study the energetic pathway for the transition between Ca^{2+} -unbound and Ca^{2+} -bound forms.^{15–18} Simulations have also incorporated cTnI into the system to elucidate how other subunits in the cTn complex can affect the dynamics of cTnC.^{19–24} Computer-aided drug discovery has also proven effective in helping identify potential therapeutics for heart disease by targeting the cTn complex.^{25–29}

How mutations located in regions not involved in either the coordination of Ca^{2+} to site II or the cTnI_{SP}–cTnC_{HP} binding event can alter the overall dynamics of muscle contraction remains underexplored. One hypothesis on how some of these mutations, particularly those in the cTnI inhibitory peptide (cTnI_{IP}), may affect troponin function was that they alter the availability, or effective concentration, of the cTnI_{SP} to the cTnC_{HP}. Our lab previously explored this hypothesis by determining the volume sampled by the cTnI_{SP} during MD simulations. We saw no significant difference in cTnI_{SP} effective concentration between wild type (WT) and mutations in the cTnI inhibitory peptide (residues 137–148),¹⁹ leading us to search for other computational methods to explore the effects of mutations on cTnC dynamics.

Another interesting interaction that has been, at least computationally, underexplored is the binding of the cTnI_{SP} to the cTnC_{HP} region. Experimentally, this has been studied

before by Li et al.³⁰ in an NMR kinetic study of cTnI_{147–163} titrated into a Ca^{2+} -bound NcTnC. The authors estimated a binding K_D of 154 μM .³⁰ Another experiment by Tikunova et al. estimated the binding affinity to be 200 nM using IAANS fluorescence,³¹ measuring an almost three orders of magnitude difference as compared to the first study. The difference is likely explained by the different sequences of cTnI used, where the NMR experiment used residues cTnI_{147–163}, and the fluorescence experiment used cTnI_{128–180}. Because C-terminal residues of cTnI are largely structural,⁶ the additional 17 residues located on the C-terminal end of the cTnI_{SP} may significantly increase the binding affinity of cTnI_{SP} to the cTnC_{HP}. In order for the cTnI_{SP} to bind to the cTnC_{HP}, cTnC first needs to undergo an opening event. Although this event is difficult to measure experimentally,³² it has been studied computationally before by Bowman et al. using a model of just the N-terminal region of cTnC.¹⁶

We therefore set out to estimate the free energy associated with the cTnI_{SP}–cTnC_{HP} binding and the opening of the cTnC_{HP} using a combination of MD-based simulations and free energy methods. To obtain a trajectory of conformations between the Ca^{2+} -unbound, cTnI_{SP}-unbound structure and the Ca^{2+} -bound, cTnI_{SP}-bound structure, we employed targeted molecular dynamics (TMD). This specific type of MD simulation can steer a starting model toward a target model by applying an additional potential energy function.³³ The implementation of this MD method has been shown recently to help elucidate transition pathways of other proteins.^{34,35} Using the conformations along this transition, we performed US to determine the free energy associated with two separate events: opening of the cTnC_{HP} and cTnI_{SP} binding to cTnC_{HP}. We also applied this same method to mutations known to be linked with HCM or RCM, one mutation that was experimentally designed to increase Ca^{2+} sensitivity of cTnC, and the phosphorylation of cTnI Thr143. Given that Ca^{2+} binding causes opening of the cTnC_{HP}, we hypothesized that mutations linked with HCM or RCM and/or known to experimentally increase Ca^{2+} sensitivity would lower the free energy associated with the cTnC_{HP} opening event. For the cTnI_{SP}–cTnC_{HP} interaction, we hypothesized that these same mutations may cause the free energy of cTnI_{SP}–cTnC_{HP} binding to become less favorable due to the mutations potentially disturbing important residue–residue contacts necessary for binding.

We were able to make two important observations from these simulations. First, our TMD method followed by US was able to calculate values for both the cTnC_{HP} opening and cTnI_{SP} binding free energies in agreement with previous

experimental and computational studies. This agreement allowed us to conclude that our method is a reliable way to study the free energy profile of the transition between Ca^{2+} -unbound and Ca^{2+} -bound conformations of cTn. Second, we saw a general slight decrease in free energy of the cTn_{CHP} opening upon introduction of mutations linked with HCM and RCM, as well as the Ca^{2+} -sensitizing mutation L48Q. However, we did not observe a general weakening or strengthening of the cTnI_{SP}–cTn_{CHP} binding upon introduction of these same mutations.

METHODS

Model Preparation for TMD. To run TMD, a start and end point for the cTn complex simulations needed to be obtained. For the starting model, chains a, b, and c from PDB entry 6KN7⁵ were extracted (Figure 1A). This model provides atom positions for all three subunits of cTn in the Ca^{2+} -unbound, cTnI_{SP}-unbound form in the context of the thin filament. For the end point of the TMD simulation, chains a, b, and c from PDB entry 6KN8⁵ were extracted (Figure 1B). This model provides atom positions for all three subunits of cTn in the Ca^{2+} -bound, cTnI_{SP}-bound form in the context of the thin filament. Since TMD requires identical atoms for the start and end models, the following residues for both extracted models were used in the simulations: cTnC residues 2–161, cTnI residues 41–166, and cTnT residues 199–272. Neither of these two PDBs provided information on the position of Ca^{2+} ions, so Ca^{2+} was added to sites II, III, and IV of the cTnC in both models using AutoDock Vina.³⁶ Although the 6KN7 model represents a Ca^{2+} -unbound, cTnI_{SP}-unbound form of the cTn complex, Ca^{2+} binding initiates the conformational change to the cTnI_{SP}-bound form of the cTn complex, making the addition of Ca^{2+} to site II of the start model necessary to simulate the conformational change appropriately.

TMD Optimization and Simulations. For the TMD simulations, the structures were solvated with explicit TIP3W water molecules and NaCl counterions were added to neutralize the entire system and establish a final salt concentration of 150 mM. The system was minimized over two sequential 10,000 minimization steps to optimize the positions of the solvent molecules around the protein and to subsequently relax the positions of residue side-chain atoms. The system was then slowly heated up to 310 K over an initial equilibration of 190,000 steps, followed by a final equilibration of 10,000 steps, to bring the system into the final production run conditions. Pressure was controlled at atmospheric conditions during equilibration simulations using Langevin piston pressure control. All preparation steps and TMD simulations were conducted using NAMD 2.13³⁷ and a 2 femtosecond timestep. Parameters for all atoms within the simulation were determined from the Charmm36 force field.³⁸ Although previous studies have detailed difficulties of modeling Ca^{2+} ion interactions and Ca^{2+} -binding free energies in MD simulations involving biomolecules, the Charmm36 force field has been shown as a reliable force field option for producing experimental-like configurations of proteins in simulations where Ca^{2+} is bound to an EF-hand protein (i.e., troponin).³⁹ The TMD simulations were conducted using an NPT ensemble at 310 K with Langevin temperature and pressure dampening. All bonds with hydrogens were constrained using the ShakeH algorithm, which allowed for a 2 femtosecond timestep, with structures being saved every 2 picoseconds.

Three independent TMD simulations were performed for 60 ns using a force constant of $k = 200 \text{ kcal/mol/\AA}^2$. Technical details of TMD simulations, the process for determining optimal parameters for the cTn system, and post-simulation processing and analysis of the trajectories are described in the Supporting Information. For each frame in these trajectories, two values were determined: the interhelical distance between cTnC helices A and B and the distance between the cTnI_{SP} and the cTnC hydrophobic patch. Interhelical distance was used as a proxy for interhelical angle to evaluate the degree of openness of the hydrophobic patch by measuring the distance between residues 14 and 48 of cTnC. The position of residues 14 and 48 in each frame, respectively, was determined by averaging the position of the residues' N, C, and CA atoms. The distance between the cTnI_{SP} (residues 149–164) and the cTn_{CHP} (residues 20, 23, 24, 26, 27, 36, 41, 44, 48, 57, 60, 77, 80, and 81) was determined by calculating the average position of the CA atoms in the collective group of residues for each region and measuring the distance between the two centers of mass. The trial that exhibited the most sampling close to the line of regression was selected as the representative trajectory from which frames were extracted to run US.

Umbrella Sampling. Specific frames from the representative TMD trajectories were selected as windows for US simulations (Table S1). The timestep for these simulations was set to 1 fs. For each US window, two independent sets of simulations were performed: one set of simulations with the cTn_{CHP} interhelical distance as a collective variable and another independent set of simulations with the cTnI_{SP}–cTn_{CHP} distance as a collective variable. The variables were defined as shown in Table S1, and a harmonic force constant of 5 kcal/mol was applied. Interhelical distance was restrained by evaluating the distance between the average position of atoms N, C, and CA in residues 14 and 48 of cTnC. The cTnI_{SP}–cTn_{CHP} distance was restrained by evaluating the average position of the CA atoms in the collective group of residues for each region (see the TMD section above). Each of the 27 windows was then run for 8 ns in 3 independent simulations for each of the collective variables, at 310 K under an NPT ensemble, similar to the TMD simulations.

Simulations that used cTn structures with cardiomyopathic mutations, phosphorylation, or Ca^{2+} -sensitizing mutations were created by modifying the windows described above. We selected all cTn mutations in the cTnI_{IP}, cTnI_{SP}, and NcTnC regions that have been previously linked with cardiomyopathy.² For many of those mutations, unfortunately, there was no experimental data available on the free energy of opening the cTn_{CHP} or the cTnI_{IP} binding affinity. The designed L48Q mutation in the NcTnC region was included due to extensive experimental research revealing that the mutation has a Ca^{2+} -sensitizing effect.^{40,41} The post-translational modification of T143 has been shown to physiologically occur and has been suggested to cause a Ca^{2+} -sensitizing effect.⁴² For cardiomyopathic or Ca^{2+} -sensitizing mutations, the mutations were introduced to each window using the Mutagenesis Wizard function available in PyMOL.⁴³ Mutations introduced to cTnC include A8V, L29Q, A31S, L48Q, and C84Y. Mutations introduced to cTnI include R141Q, L144P, L144Q, R145G, R145Q, R145W, A157V, R162P, R162Q, and R162W. The cTnI Thr143-phosphorylated (T143p) windows were created using the PyTMs plug-in available in PyMOL.⁴⁴ Each of the modified windows was also run for 8 ns in 3 independent simulations, at 310 K under an NPT ensemble.

WHAM Free Energy Estimation. For each frame in each US trajectory, the values of the applied collective variable were evaluated. The Weighted Histogram Analysis Method (WHAM)⁴⁵ was used to create a free energy profile based on the individual collective variable evaluated. This allowed us to estimate free energy profiles of the cTnC_{HP} opening transition coordinate and cTnI_{SP} association coordinate, respectively, without bias of the other collective variable restraining the system during simulation.

The final reported values for the free energy attributed to each collective variable were determined by averaging the data from the three independent simulations for each system, with errors being reported as the sample standard deviation between the three trials. The minimum and maximum values provided to WHAM for the interhelical distance collective variable were 14.5 and 28.5 Å, respectively. The minimum and maximum values provided to WHAM for the cTnI_{SP}–cTnC_{HP} distance collective variable were 9.6 and 16.2 Å, respectively. The histogram analysis for both profiles was divided into 27 bins, one for each of the windows simulated, with the center of the bin being defined as the energy minimum for the window. A tolerance of 0.00001 was applied, and a force constant of 10 kcal/mol was used after properly adjusting the 5 kcal/mol spring constant applied in the Charmm36 force field. Adjustment of the spring constant for implementing WHAM was necessary because the Charmm36 force field does not include a $2/1$ when specifying spring constants for the restraint terms, yet the code for WHAM does. This force constant adjustment is described in “An implementation of WHAM: the Weighted Histogram Analysis Method” by Alan Grossfield.⁴⁵

RESULTS

TMD Simulations Generate a Transition Pathway for Umbrella Sampling. TMD simulations were used to obtain a trajectory of the transition from a Ca²⁺-unbound, cTnI_{SP}-unbound cTn complex to a Ca²⁺-bound, cTnI_{SP}-bound cTn complex. We optimized two variables in the potential energy function for the TMD term: the force constant, k (kcal/mol/Å²), and the length of the simulation, t . TMD simulations were run for five separate force constants, at increasing simulation lengths. For each simulation, the interhelical angle between cTnC helices A and B for all frames captured from the trajectories was determined. TMD trajectories run for $t = 20$ ns and $t = 40$ ns exhibited abnormally high interhelical angles for the lowest three force constants ($k = 50$ kcal/mol/Å², $k = 100$ kcal/mol/Å², and $k = 200$ kcal/mol/Å²), while higher force constants ($k = 500$ kcal/mol/Å² and $k = 1000$ kcal/mol/Å²) led to a rapid cTnC_{HP} opening (Figure S1). Additionally, visual inspection for all these simulations revealed denaturation of the cTnI_{SP} into a disordered region before interaction with the cTnC_{HP}. For the $t = 60$ ns simulations, the trajectories with $k = 50$ kcal/mol/Å² and $k = 100$ kcal/mol/Å² force constants again experienced unnaturally high interhelical angles. The $k = 200$ kcal/mol/Å² and $k = 500$ kcal/mol/Å² trajectories exhibited sufficient sampling of both the closed and open states of the cTnC_{HP} while also sampling a smooth transition between the two (Figure 2). Visual inspection of the cTnI_{SP} region revealed that cTnI_{SP} maintained its helical nature for both force constants throughout the 60 ns simulation. We therefore chose to move forward with the parameters $t = 60$ ns and $k = 200$ kcal/mol/Å² to avoid denaturation of parts of the cTn complex. Simulations ran for $t = 80$ ns and $t = 100$ ns across all force constants also sampled the anticipated ranges of

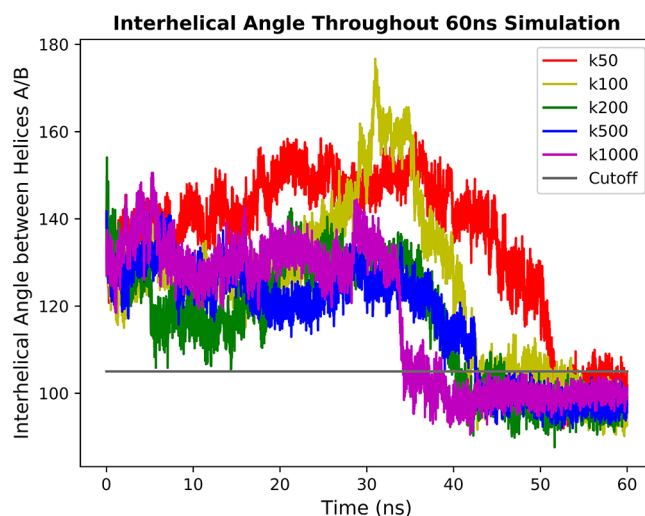


Figure 2. Interhelical angle throughout the 60 ns TMD simulations. Time of the simulation vs interhelical angle of NcTnC helices A and B. The gray “cutoff” line at the 105° mark was added to differentiate between open and closed conformations. Force constants are colored: red ($k = 50$ kcal/mol/Å²), yellow ($k = 100$ kcal/mol/Å²), green ($k = 200$ kcal/mol/Å²), blue ($k = 500$ kcal/mol/Å²), and purple ($k = 1000$ kcal/mol/Å²).

interhelical angles and exhibited gradual cTnC_{HP} opening events, but we decided to use the shorter $t = 60$ ns simulations for production runs.

Three independent trials of TMD of the WT cTn complex were run using these parameters ($k = 200$ kcal/mol/Å² and simulation length of 60 ns), and the sampling along the cTnC_{HP} patch opening coordinate was investigated. The correlation between the interhelical angle and the interhelical distance for helices A/B can be seen in Figure S2. Furthermore, the data for each frame in the trajectories was analyzed with respect to the interhelical and cTnI_{SP}–cTnC_{HP} distances, with a line of regression being created along the patch opening event (Figure 3A). To obtain full coverage of the transition coordinate between the closed and open hydrophobic patch, target values for the interhelical distance were set every 0.5 Å between 15.0 and 28.0 Å, creating 27 target interhelical distances. This corresponded to interhelical angles ranging from approximately 95 to 130°. Each of these values was paired according to the regression with a target cTnI_{SP}–cTnC_{HP} distance to obtain full coverage of the transition coordinate between an unbound cTnI_{SP} and a bound cTnI_{SP}. This resulted in 27 windows with pairs of target values, ranging from window 1 with a closed cTnC_{HP} and unbound cTnI_{SP} (interhelical distance = 15.0 Å, cTnI_{SP}–cTnC_{HP} distance = 15.7 Å) to window 27 with an open cTnC_{HP} bound to the cTnI_{SP} (28.0 Å, 10.3 Å) (Figure 3B). Figure 3C–E shows the starting conformation of the cTn complex for windows 1, 14, and 27, respectively. Window 1 shows the cTnI_{SP} near a closed cTnC_{HP}, window 9 contains the cTnI_{SP} in a primed position to bind to the cTnC_{HP} (while the patch is only in a semi-open conformation), and window 27 corresponds to the cTnI_{SP} completely bound to an open cTnC_{HP}.

WT Free Energies of Patch Opening and Switch Peptide Binding Agree with Previous Experiments. For the WT system, we simulated the 27 extracted windows in triplicate for each collective variable using US and evaluated

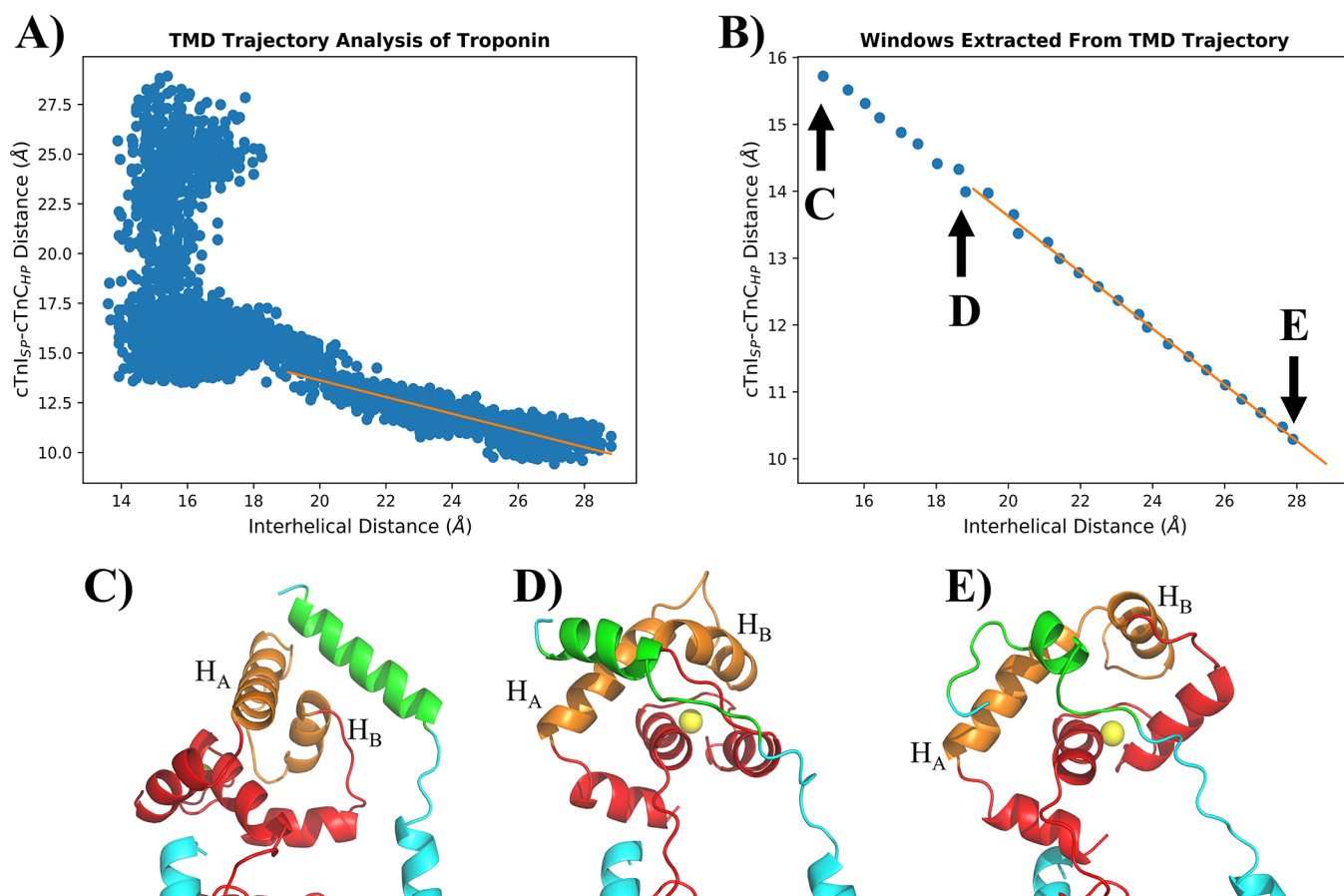


Figure 3. TMD trajectory analysis and window extraction. A) Each point represents a frame from the TMD trajectory. Line of regression was created using all data points where the interhelical distance is ≥ 19.0 Å. B) Each point represents one of the 27 windows extracted for input of US. The same line of regression from (A) is shown in orange. Simulation windows shown in (C), (D), and (E) are labeled with arrows. C) Window 1 with an interhelical distance of 15.0 Å and a cTnI_{SP}-cTnC_{HP} distance of 15.8 Å. This window has the cTnI_{SP} near a closed cTnC_{HP} and represents the start point of the transition coordinate. D) Window 9 with an interhelical distance of 19.0 Å and a cTnI_{SP}-cTnC_{HP} distance of 14.1 Å. This window has the cTnI_{SP} in a primed position to bind to the cTnC_{HP}; however, the patch is only in a semi-open conformation. E) Window 27 with an interhelical distance of 28.0 Å and a cTnI_{SP}-cTnC_{HP} distance of 10.3 Å. This window has the cTnI_{SP} completely bound to an open cTnC_{HP} and represents the end point of the transition coordinate. All regions of the cTn complex are colored identical to Figure 1. cTnC helices A and B (orange) labeled as H_A and H_B, respectively.

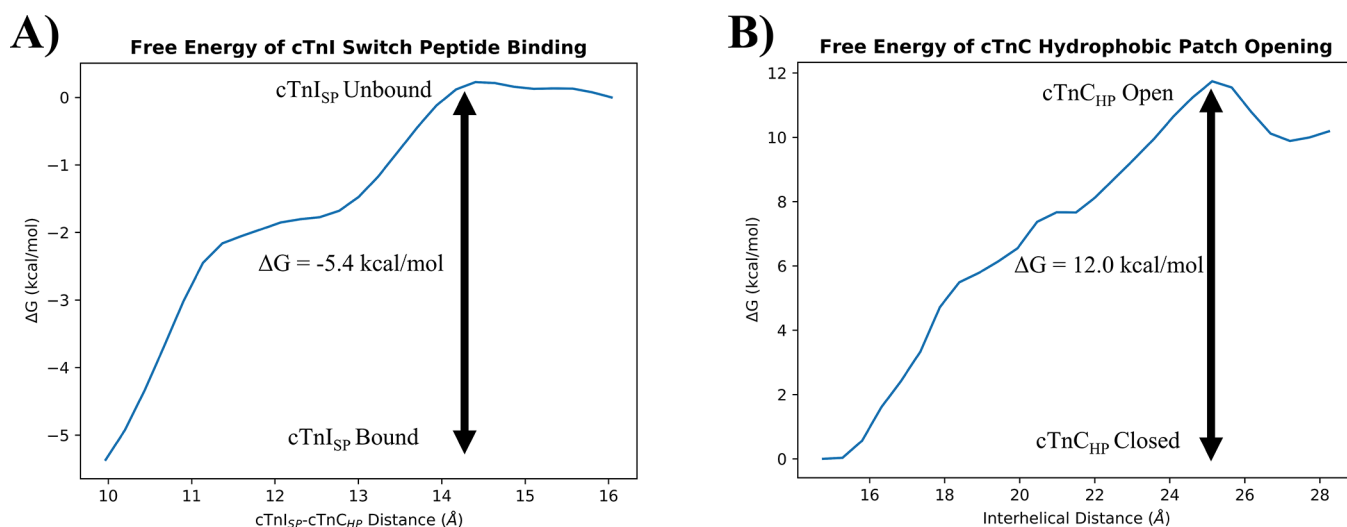


Figure 4. WHAM calculated free energy of WT transition coordinates. A) The free energy across the cTnI_{SP}-cTnC_{HP} binding coordinate for the WT cTn model. Binding occurs right (window 7) to left (window 27) as the distance between the two entities decreases. B) The free energy across the cTnC_{HP} opening coordinate. The cTnC_{HP} opens from left (window 1) to right (window 27).

the results with WHAM. We created a free energy profile considering each collective variable independently (Figure 4). Analysis of the US simulations revealed that the evenly spaced windows exhibited sufficient overlap and that we were able to sample the entire reaction coordinate (Figure S3). Evaluating the convergence of the free energy calculated by WHAM as a function of simulation time revealed that 8 ns was sufficient for sampling the windows of both collective variables (Figure S4). The free energy estimated for the cTnI_{SP} binding event only fluctuated by about 0.1 kcal/mol after 5 ns (Figure S4A), and the cTnC_{HP} opening free energy only changed by 0.3 kcal/mol after 5 ns (Figure S4B). The estimated free energy of the cTnI_{SP}–cTnC_{HP} binding event was then determined to be -5.4 ± 0.5 kcal/mol. Figure 4A shows the calculated free energy per window as a function of the distance between the cTnI_{SP} and cTnC_{HP}. The free energy was measured from the peak of the free energy curve in window 7 (the unbound state; at a cTnI_{SP}–cTnC_{HP} distance of about 14.5 Å) to the free energy estimation in the last window (the bound state; at a cTnI_{SP}–cTnC_{HP} distance of about 10.3 Å; window 27). This was done because the cTnI_{SP} was not in proximity to the cTnC_{HP} binding site until window 7, when the patch opening reached a semi-open conformation suitable for cTnI_{SP}–cTnC_{HP} binding to commence.

An NMR kinetic study of cTnI_{147–163} titrated to a Ca²⁺-bound NcTnC estimated a binding K_D of 154 ± 10 μM.³⁰ This experimental study is the most comparable to our simulation data because a highly similar region for the cTnI_{SP} was used. At $T = 303.15$ K (30 °C), this data corresponds to an experimentally determined free energy of -5.3 ± 0.1 kcal/mol, a value in perfect agreement with our computational value of -5.4 ± 0.5 kcal/mol. Other experimental studies explored the binding of longer cTnI segments to cTnC. However, since residues located C-terminally of the cTnI_{SP} were not resolved in the 6KN8 PDB structure, we were unable to test the effect of these residues on the switch peptide binding affinity. Given this small deviation, we hypothesized that our method of using TMD to produce windows representative of a transition coordinate, in combination with US to measure the free energy along that coordinate, can reliably reproduce experimentally determined free energy values of events associated with the cTn conformational transition. Applying the same strategy to the cTnC_{HP} opening transition coordinate, we estimated the free energy of the opening event to be 12.0 ± 1.3 kcal/mol. The WHAM determined free energy as a function of the opening coordinate can be seen in Figure 4B. Although the free energy of this event has not been studied experimentally, the result agrees with previous computational simulations that measured the free energy of cTnC_{HP} opening in NcTnC to be 13.8 ± 2.2 kcal/mol using a combination of steered MD and US.¹⁶

To test the effect of the cTnI_{SP} getting incrementally closer to a closed hydrophobic patch, we extracted 11 windows from our TMD simulations (each at a cTnC_{HP} interhelical distance of 15 Å), with a decreasing cTnI_{SP}–cTnC_{HP} distance between 25 and 20 Å (step size 0.5 Å). Between these windows, we observed only a small change in free energy (~ 1.8 kcal/mol) as the cTnI_{SP} shifted its position closer to the cTnC_{HP} (Figure S5). We would not expect this small difference to significantly affect the reported results.

HCM and RCM Mutations Generally Decreased cTnC_{HP} Opening Free Energy. After developing a reliable method to measure the free energy of the cTnC_{HP} opening and

cTnI_{SP}–cTnC_{HP} binding events, we subsequently introduced mutations and post-translational modifications in cTn to determine how these changes affect the free energies of patch opening and switch peptide binding. We tested 12 mutations that have been linked with HCM, two mutations linked with RCM, one designed mutation that increases Ca²⁺ sensitivity without causing disease, and phosphorylation of cTnI Thr143. All modifications tested were located in the NcTnC region or between residues 141–162 of cTnI. Data for the estimated free energy of patch opening and switch peptide binding for all 16 modified systems can be seen in Table 1.

Table 1. Free Energy of cTnC_{HP} Opening and cTnI_{SP} Binding Events for All Models. All data shown in units of kcal/mol.

Model	Disease	ΔG cTnC _{HP} Opening	ΔG cTnI _{SP} Binding
Wild-Type	---	12.0 ± 1.3	-5.4 ± 0.5
cTnC A8V	HCM	11.6 ± 1.7	-3.9 ± 1.3
cTnC L29Q	HCM	13.1 ± 1.8	-7.2 ± 1.2
cTnC A31S	HCM	11.4 ± 1.0	-4.3 ± 1.0
cTnC L48Q	---	4.8 ± 1.5	-8.0 ± 0.2
cTnC C84Y	HCM	8.4 ± 1.1	-6.4 ± 0.6
cTnI R141Q	HCM	13.6 ± 0.7	-4.3 ± 0.7
cTnI L144P	HCM	12.3 ± 2.2	-3.6 ± 1.5
cTnI R145G	HCM	12.5 ± 1.0	-5.5 ± 1.0
cTnI R145Q	HCM	13.5 ± 2.5	-7.2 ± 0.6
cTnI A157V	HCM	12.5 ± 1.9	-5.6 ± 2.2
cTnI R162P	HCM	12.6 ± 2.5	-3.5 ± 1.1
cTnI R162Q	HCM	11.4 ± 1.8	-4.3 ± 0.9
cTnI R162W	HCM	11.8 ± 0.9	-4.4 ± 1.3
cTnI L144Q	RCM	11.7 ± 2.5	-4.7 ± 0.5
cTnI R145W	RCM	10.4 ± 1.8	-5.6 ± 1.0
cTnI T143p	---	15.6 ± 0.9	-4.6 ± 1.5

Two of the mutations (cTnC L48Q and C84Y) exhibited a significant decrease in the free energy of cTnC_{HP} opening (Figure 5A), while the other 13 mutations caused no significant change in the free energy.

Two mutations had a drastic effect on the cTnC_{HP} opening: L48Q (4.8 kcal/mol) and C84Y (8.4 kcal/mol). Interestingly, the mutation that caused the biggest change in free energy of this opening event was L48Q, a designed cTnC Ca²⁺-sensitizing mutation that has not been linked with disease.^{40,41} This mutation also caused the most favorable cTnI_{SP}–cTnC_{HP} interaction (-8.0 kcal/mol) among all mutations studied (see the next section). The only cardiomyopathic mutation that we observed having a significant effect on the cTnC_{HP} opening was cTnC C84Y. This mutation has been shown experimentally to increase Ca²⁺ sensitivity of force generation in skinned cardiac muscle fibers.⁴⁶

Mutations in the cTnI inhibitory peptide and cTnI_{SP} regions did not affect the free energy of cTnC_{HP} opening significantly. Prior experiments have suggested that cTnI mutations that cause a more severe increase in Ca²⁺ sensitivity lead to RCM, whereas mutations that cause a less severe increase in Ca²⁺ sensitivity lead to HCM.⁴⁷ Our data would support this theory since mutations linked with RCM in the cTnI inhibitory peptide and cTnI_{SP} regions (L144Q and R145W) generally had a larger effect on the cTnC_{HP} opening free energy when compared to HCM mutations in the same region. Currently, no RCM mutations have been identified in cTnC,^{2,48} so we were unable to test this trend among the cTnC mutations.

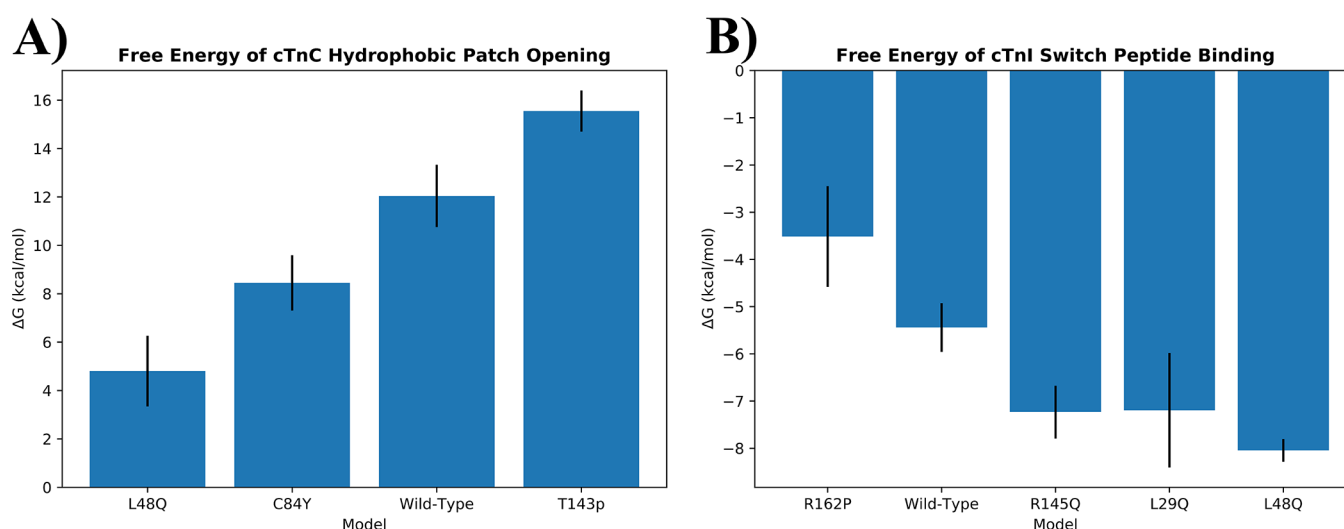


Figure 5. Free energy of cTnC_{HHP} opening and cTnI_{SP} binding events for all simulated systems that caused a significant impact. A) WHAM calculated ΔG for the cTnC_{HHP} opening event for cTn systems ordered based on average ΔG . B) WHAM calculated ΔG for the cTnI_{SP} binding event for cTn systems ordered based on average ΔG . The standard deviations for each system are shown as black bars.

No Observed Trend in cTnI_{SP}–cTnC_{HHP} Binding Free Energy. We were unable to observe any consistent trend in the free energy of cTnI_{SP}–cTnC_{HHP} binding. Three mutations (cTnC L29Q, cTnC L48Q, and cTnI R145Q) caused a significantly more favorable interaction, one mutation (cTnI R162P) caused a less favorable interaction, and the other eleven mutations had no measurable effect on cTnI_{SP}–cTnC_{HHP} binding (Figure 5B and Table 1). The effect of cTn mutations on cTnI_{SP}–cTnC_{HHP} binding has been scarcely studied before. A study by Tikunova et al. that estimated the binding affinity of cTnI_{128–180} to cTnC in the presence of Ca²⁺-sensitizing, non-cardiomyopathic mutations observed a slight increase in K_D when compared to the WT.³¹ We would therefore hypothesize to see a less favorable interaction between the cTnI_{SP} and the cTnC_{HHP} given mutations linked with HCM or RCM or those that have been experimentally shown to cause a Ca²⁺-sensitizing effect. Although we did observe slightly less favorable interactions in most of the mutated models, only one model produced a significantly less favorable interaction. For the one Ca²⁺-sensitizing, non-cardiomyopathic mutation that was studied (L48Q), we observed a significantly more favorable cTnI_{SP}–cTnC_{HHP} interaction (–8.0 kcal/mol) as compared to WT. Interestingly, the other cTnC mutation that caused a more favorable interaction (L29Q) was also a leucine to glutamine substitution.

Thr143 Phosphorylation May Increase cTnC_{HHP} Opening Free Energy. PKA-mediated phosphorylation of cTnI has been shown as a mechanism for increasing the heart rate under stress by allowing the heart muscle to contract at a faster rate as a consequence of a decrease in Ca²⁺ sensitivity from phosphorylation of cTnI Ser 22/23.⁴⁹ However, studies of the PKC-mediated phosphorylation of cTnI Thr143 have actually shown the adverse effect, namely, that phosphorylation of this site may cause an increase in Ca²⁺ sensitivity in the myofilament.⁴² Our phosphorylated model of cTnI Thr143 (cTnI T143p) slightly increased the cTnC_{HHP} opening free energy (15.6 kcal/mol) while having no effect on the free energy of cTnI_{SP}–cTnC_{HHP} binding. Therefore, our simulations suggested that the phosphorylation of cTnI T143 might decrease calcium sensitivity through an increase in the free energy of cTnC_{HHP} opening. A possible explanation for this

apparent inconsistency with the experimentally observed increase in Ca²⁺ sensitivity was highlighted in a study by Vymětal et al.⁵⁰ The authors showed that current force fields were unable to simulate phosphorylated residues in a way that would agree with experimental observations. Therefore, the phosphorylated Thr143 in our cTn system was most likely behaving uncharacteristically, leading to results that did not corroborate experimental data.

CONCLUSIONS

In this study, we modeled the transition of cTn from the Ca²⁺-unbound to Ca²⁺-bound states by using TMD. Using the trajectory of this transition, we subsequently performed US simulations to determine the free energy associated with two separate events: opening of the cTnC_{HHP} and cTnI_{SP} binding to cTnC_{HHP}. The results obtained for the cTnI_{SP}–cTnC_{HHP} interaction in the WT model agreed with previous experimental results,³⁰ while our cTnC_{HHP} opening free energy estimation agreed with previous computational efforts.¹⁶ This excellent agreement for WT cTn demonstrates the important role of advanced computational methods in quantitatively studying cTn dynamics and muscle contraction. The methodology described here, using TMD followed by US, could also be applied to other protein systems given the availability of structures in two different states.

We further applied the methodology to cTn mutations and post-translational modifications. Our results showed that mutations linked with HCM or RCM caused a Ca²⁺-sensitizing effect, generally lowering the free energy associated with the cTnC_{HHP} opening. We observed that most of the studied mutations caused a less favorable free energy associated with cTnI_{SP}–cTnC_{HHP} binding, although this trend did not hold for all HCM and RCM mutations, with three mutations leading to significantly more favorable binding free energies for this event. Our method could impact the future of computational studies on this system by predicting the consequences of unknown mutations on cTn energetics. Additionally, this method could prove helpful in rational protein design. Overall, we showed that our protocol of utilizing US to measure the free energy of transitions sampled by TMD is a reliable method for studying the energetics of the cTn complex.

■ ASSOCIATED CONTENT

Data Availability Statement

The wild-type cTn protein structures (6KN7, 6KN8) were available in the Protein Data Bank (PDB) at <https://www.rcsb.org/>. All mutations were created using the wild-type protein structure as the base model within the PyMOL software using the protein mutagenesis tool. Phosphorylation was created using the wild-type protein structure as the base model within the PyMOL software using the PyTMs Plug-in. All TMD and US simulations were performed within the NAMD version 2.13 framework. Weighted Histogram Analysis Method (WHAM) was performed using WHAM version 2.0.11 available at http://membrane.urmc.rochester.edu/?page_id=126.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.2c00508>.

Methods for optimization of TMD simulation parameters, values for collective variables used for each window in US, interhelical angle analysis during TMD simulations, comparison of interhelical angle vs interhelical distance, coverage of collective variables in US simulations, free energy convergence of US simulations, and free energy analysis of cTn_{SP} approaching a closed cTn_{HP} (PDF)

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Notes

The authors declare no competing financial interest.

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